
Education

University of Management and Technology

Bachelor of Science in Software Engineering. GPA: 3.7/4.0

Lahore, Pakistan

Expected Graduation: February, 2027

Relevant Coursework: Data Structures & Algorithms, Database Systems, Software Architecture, Software Engineering, Object-Oriented Programming (OOP), Computer Networks, Operating Systems

Queen Mary College

Intermediate Pre-Engineering

Lahore, Pakistan

2020-2022

Grade: A

Projects

1. AI Job Market Analysis & Salary Prediction

- Analyzed 15,000 AI/ML job postings to identify skill demand trends and salary patterns.
- Applied KMeans clustering to discover job role groups based on required skills.
- Built a Random Forest regression model ($R^2 \approx 0.58$) to predict salaries and visualize insights.

2. Django User Authentication & Profile Management Web Application

- Engineered a secure authentication and profile management system using Django's built-in authentication framework.
- Implemented user registration, login/logout, and session-based access control.
- Designed a custom Profile model (One-to-One with User) for extended user data.
- Developed profile update functionality with image upload and server-side validation.
- Applied server-side validation, template inheritance, and modular project organization for maintainability.

3. Desktop Application – PyQt5 & Qt Designer

- Developed a GUI-based desktop application using Qt Designer for frontend design and Python (PyQt) for backend logic.
- Implemented event-driven programming and modular architecture separating UI and business logic.
- Integrated persistent data handling using file/database storage.
- Applied object-oriented programming principles for structured and maintainable code.

4. Dynamic CV Builder – HTML, CSS, JavaScript

- Built a client-side web application enabling users to generate structured CV templates through form-based input.
- Implemented real-time DOM manipulation for dynamic CV rendering.
- Designed responsive layouts using HTML5 and CSS3.
- Applied client-side validation for accurate and structured data handling.

5. Event Management System – Java

- Developed an event scheduling and registration platform using 5 design patterns.
- Increased modularity and reduced code duplication by 25%, enhancing long-term maintainability.

6. Portfolio Website

- Built a responsive portfolio using HTML, CSS, and JavaScript to showcase academic and personal projects.

7. E-commerce Database Design – MySQL

- Designed an optimized relational database schema with 8 tables using 3NF normalization and relational constraints.

Certificates

Machine Learning with Python – freeCodeCamp

- Learned core ML concepts including supervised learning, regression, classification, and model evaluation using Python and Scikit-learn.
- Applied data preprocessing, feature engineering, and model training through practical exercises.

Responsive Web Design – freeCodeCamp

- Completed a structured program covering HTML5, CSS3, and responsive layout techniques.
- Built 20+ practice projects to strengthen front-end development skills.

Web Design for Web Developers – Udemy

- Studied modern UI principles, color theory, and responsive design practices.
- Applied these concepts to build and refine multiple sample web pages.

Technical skills

- Programming Languages: Python, Java, C++, JavaScript, SQL, HTML5, CSS3
- Frameworks & Libraries: Django, React, Node.js, PyQt5, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
- Web Scraping: Selenium
- Tools & Version Control: Git, GitHub, VS Code
- Other Tools: Kali Linux, Cisco Packet Tracer